Week 2 – Strings + Recursion

Day 8 – String Basics & StringBuilder

1️⃣ String Basics (Java)

String kya hota hai?

String = characters ka sequence

String str = "Chandan";

🔹 Important Points

String immutable hota hai
👉 Matlab change nahi hota (new object ban jata hai)

🔹 Common String Methods

1. Length

```
public class StringLength {
    public static void main(String[] args) {

        String str = "Hello";

        System.out.println(str.length()); // 5
    }
}
```

2. CharAt

```
public class CharAtExample {
    public static void main(String[] args) {

        String str = "Hello";

        System.out.println(str.charAt(1)); // e
    }
}
```

3. Compare Strings

```
public class CompareString {
    public static void main(String[] args) {

        String a = "Hello";
        String b = "Hello";

        System.out.println(a.equals(b)); // true
    }
}
```

4. Substring

```
public class SubstringExample {
    public static void main(String[] args) {

        String str = "Chandan";

        System.out.println(str.substring(0,4)); // Chan
    }
}
```

2 Traverse String

```
public class TraverseString {
    public static void main(String[] args) {

        String str = "Java";

        for(int i = 0; i < str.length(); i++){
            System.out.println(str.charAt(i));
        }
    }
}
```

3 Reverse String (Important)

```
public class ReverseString {
    public static void main(String[] args) {

        String str = "hello";
        String rev = "";

        for(int i = str.length()-1; i >= 0; i--){
            rev = rev + str.charAt(i);
        }

        System.out.println("Reversed = " + rev);
    }
}
```

⚠️ Problem:

Yeh slow hai ($O(n^2)$) kyunki string immutable hai

4️⃣ StringBuilder (Important)

StringBuilder kya hota hai?

Mutable (change ho sakta hai)

Fast (efficient for modification)

- Create StringBuilder

```
StringBuilder sb = new StringBuilder("Hello");
```

- Important Methods

1. Append

```
 sb.append(" World");
System.out.println(sb); // Hello World
```

2. CharAt

```
 System.out.println(sb.charAt(1));
```

3. SetCharAt

```
 sb.setCharAt(0, 'Y');
System.out.println(sb); // Yello
```

4. Insert

```
 sb.insert(2, 'X');
```

---

5. Delete

```
 sb.delete(1,3);
```

5️⃣ Reverse String using StringBuilder (BEST)

```
public class ReverseUsingSB {
    public static void main(String[] args) {

        String str = "hello";

        StringBuilder sb = new StringBuilder(str);

        sb.reverse();

        System.out.println("Reversed = " + sb);
    }
}
```

Time Complexity: O(n) ✅

Problem 1

Check karo string palindrome hai ya nahi

```
package Day_8;

public class CheckPalidrome {
    public static void main(String[] args) {

        StringBuilder sb = new StringBuilder("mam");

        // original copy save karo
        String original = sb.toString();

        // reverse karo
        sb.reverse();

        if (original.equals(sb.toString())) {
            System.out.println("Palindrome");
        } else {
            System.out.println("Not Palindrome");
        }
    }
}

```

Problem 2

Count vowels in string

```
 1. package Day_8;
 2.
 3. public class CountVowel {
 4.     public static void main(String[] args) {
 5.
 6.         String str = "Chandan Kumar";
 7.         int count = 0;
 8.
 9.         for(int i = 0; i < str.length(); i++){
10.
11.             char ch = str.charAt(i);
12.
13.             if(ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u' ||
14.                ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' || ch == 'U'){
15.
16.                 count++;
17.             }
18.         }
19.
20.         System.out.println("Number of vowels: " + count);
21.     }
22. }
23.
```

Problem 3

String me space remove karo

1️⃣ Method 1 (Replace Method – Easy & Best)

```
public class RemoveSpace {
   public static void main(String[] args) {

       String str = "Chandan Kumar Thakur";

       String result = str.replace(" ", "");

       System.out.println(result);
   }
}
```

✅ Output

ChandanKumarThakur

👉 Ye **sabse easy aur recommended method hai**

2 Method 2 (Using Loop)

```
public class RemoveSpace {
    public static void main(String[] args) {

        String str = "Chandan Kumar Thakur";
        String result = "";

        for(int i = 0; i < str.length(); i++){

            if(str.charAt(i) != ' '){
                result = result + str.charAt(i);
            }
        }

        System.out.println(result);
    }
}
```

3 Method 3 (Using StringBuilder – Fast)

```
public class RemoveSpace {
    public static void main(String[] args) {

        String str = "Chandan Kumar Thakur";
        StringBuilder sb = new StringBuilder();

        for(int i = 0; i < str.length(); i++){

            if(str.charAt(i) != ' '){
                sb.append(str.charAt(i));
            }
        }

        System.out.println(sb.toString());
    }
}
```

🧠 Concept

Method	Speed	Use
replace()	Fast	Best
Loop	Medium	Logic samajhne ke liye
StringBuilder	Fastest	Large data